Thermodynamics and Statistical Physics

9th Exercise Class

Maxwell–Boltzmann Distribution, Mean Free Path Approximation, Transport Processes

Exercises for the Class

1. For a gas obeying the Maxwell-Boltzmann distribution:

   (a) Derive the most probable speed $v_{\mathrm{mp}} = = \sqrt{\frac{2k_{\mathrm{B}}T}{m}}$, mean speed $\langle v\rangle = \sqrt{\frac{8k_{\mathrm{B}}T}{\pi m}}$, and root-mean-square speed $v_{\mathrm{rms}} = \sqrt{\frac{3k_{\mathrm{B}}T}{m}}$.

   (b) Sketch the speed distribution, labeling all three characteristic velocities. Explain why $v_{\mathrm{mp}} < \langle v\rangle < v_{\mathrm{rms}}$.

2. Using the Maxwell-Boltzmann velocity distribution function $f(v)$, derive the corresponding energy distribution function $f(\varepsilon)$ for the kinetic energy $\varepsilon = \frac{1}{2}mv^2$.

   (a) What is the most probable kinetic energy?

   (b) What is the average kinetic energy?

3. Consider nitrogen gas (N_2) with molecular diameter $d = 0.37\,\mathrm{nm}$:

   (a) Derive the mean free path $\lambda = \frac{1}{n\sigma}$ (with $\sigma = \pi d^2$) and collision time.

   (b) At STP 100 kPa and 300 K. Calculate λ and τ. Find the average collision frequency per molecule.

   (c) How does λ change if the gas is compressed to 5 atm at constant temperature?

4. How does the diffusion coefficient D and the viscosity coefficient η of an ideal gas change when the gas volume increases by a factor of k:

   (a) at constant temperature;

   (b) at constant pressure?

5. (a) Using kinetic theory, derive the thermal conductivity κ of a monatomic gas:

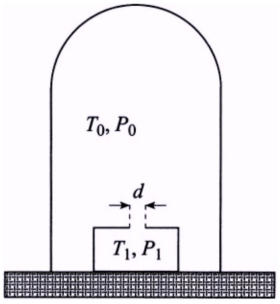
$$\kappa = \frac{1}{3}\langle v\rangle \lambda c_V,$$

   where c_V is the molar heat capacity at constant volume.

   (b) Show that κ is proportional to $T^{1/2}$ for hard-sphere molecules at fixed density.

Homework to Be Submitted

1. We place a container with a small hole of diameter d in its wall into a vacuum chamber, where the pressure is so low that the free path of the particles, λ, is much greater than d ($\lambda \gg d$). The temperature of the gas in the chamber is T_0, and its pressure is P_0. We maintain a constant temperature of $T_1 = 4T_0$ inside the container.



   Find the pressure P_1 in the container after equilibrium has been reached.

Homework for Practice

1. What fraction of molecules in a gas at a temperature T has the kinetic energy of translational motion exceeding ε_0 if $\varepsilon_0 \gg k_{\mathrm{B}}T$?

2. An ideal gas consisting of rigid diatomic molecules goes through an adiabatic process. How do the mean free path, and the number of collisions of each molecule per second depend in this process on

   (a) the volume V;

   (b) the pressure p;

   (c) the temperature T?

3. Find the temperature distribution in the space between two coaxial cylinders of radii R_1 and R_2 filled with a uniform heat conducting substance if the temperatures of the cylinders are constant and are equal to T_1 and T_2 respectively.